

Silverline House — 25-Storey Residential HRB

Fire Compartmentation Drawings

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1 Introduction

This Fire Strategy Report has been prepared to accompany the Gateway 2 application for Building Control Approval for Silverline House, a 25-storey residential higher- risk building. The strategy has been developed in accordance with Approved Document B (Fire Safety) Volumes 1 and 2, BS 9999:2017, and BS 9991:2015. The report identifies and justifies all elements of the fire safety design including means of escape, passive fire protection, active fire suppression, fire detection, and fire service access. The strategy is supported by evacuation simulations carried out in accordance with PD 7974-6 and confirmatory computational fluid dynamics (CFD) modelling of the smoke extraction system. This document index accompanies the Fire Compartmentation Drawing set SLH-FIR-100 to SLH-FIR-148.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Compartmentation Schedule

The fire compartmentation drawings show the extent of each fire compartment, fire resistance ratings of all elements, and the location of fire doors and service penetration seals. The drawings are to be read in conjunction with the Fire Strategy Report (document 2.2.1). Compartment walls between apartments are constructed in 200 mm dense aggregate blockwork or equivalent lightweight block achieving 60 minutes fire resistance in accordance with BS EN 1996-1-2 and the tabulated data in Approved Document B. Where the compartment wall coincides with the structural frame, the wall abuts the underside of the floor slab above and is sealed to prevent smoke passage. Junctions between compartment walls and the external facade are detailed to maintain compartmentation integrity at the facade interface. Product certificates and fire test evidence for all specified masonry units are included in the Construction Product Schedule accompanying this application.

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3 Drawing Register

Compartment floors are formed by the reinforced concrete flat slab construction (minimum 200 mm depth), which inherently provides 120 minutes fire resistance when designed in accordance with BS EN 1992-1-2. Where a suspended ceiling is provided below the structural slab, the ceiling system has been designed to maintain the compartment floor fire resistance rating. Penetrations through compartment floors for services (drainage, ventilation ducts, electrical conduits) are sealed with proprietary fire-stopping systems carrying third-party certification to BS EN 1366-3. The specification for these systems is contained in the Services Fire Stopping Schedule. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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